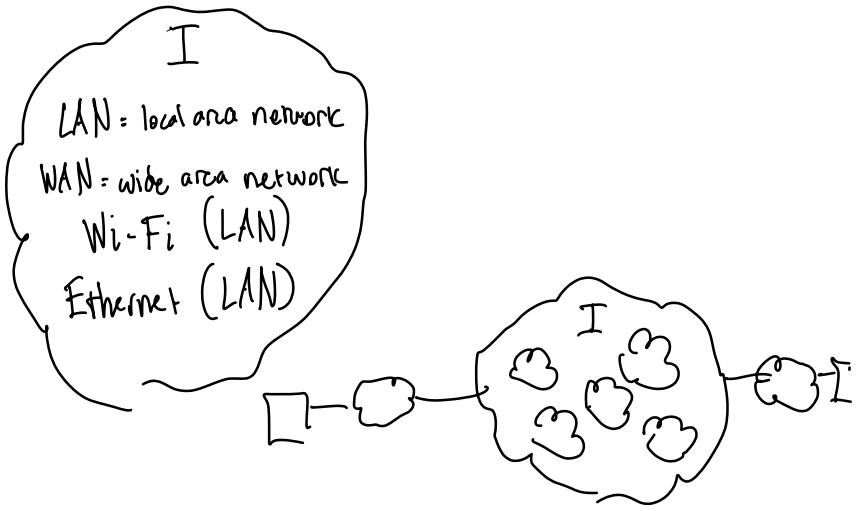
464 - Lecture 1 Notes

Cloud → Network



Physical Layer (brief)

Error Dection + Correction

Flow Control (How fast can we send and receive)

Network Protocol: defines format of information and rules needed for two devices to communicate

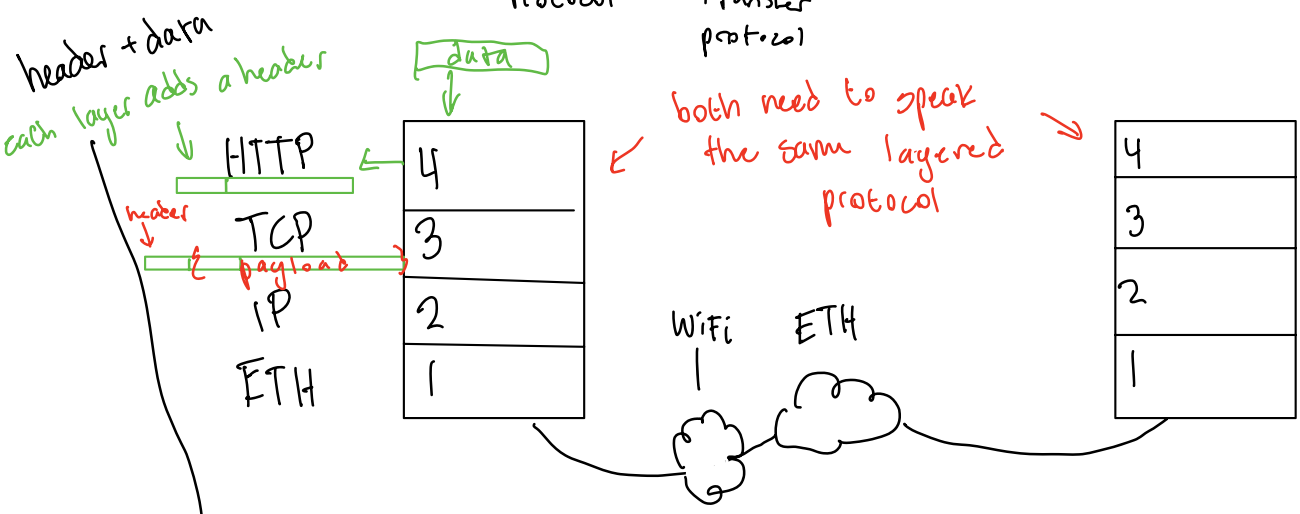
two parts:

format of data
- byte format
- language/header format

rules to be followed
- flow control
- error correction / detection

Header			
ETH	IP	TCP	HTTP
IEEE	Internet Protocol	Transmission Control Protocol	Hyper text transfer protocol

ETH = wired protocol



PDU = protocol data unit (PDU) header + data

- refers to the header + data that is being passed between layers
- this is what is created by a layer and passed to layer below

Seperation of layers:

- Encapsulation
  - the control information + data is treated as one unit by the lower level protocols
  - lower layer does NOT know what is in PDU from the layer above

layer 4 PDU: segment = payload + header

layer 3 PDU: datagram (colloquially called a packet, but that's a bad name) L3 header + payload

layer 2 PDU: frame
↑
ETH/WiFi

so what gets sent across network

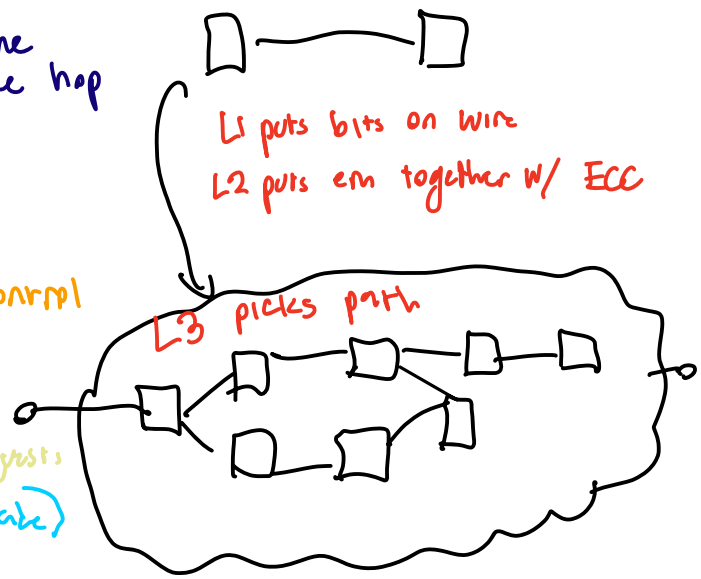
2	3	4	Payload

receiving, just strip off header as you pass it through the layer.

☆ in Wireshark, trailer gets thrown away

OSI Seven Layer Model

1. · Physical Layer (No PDU, purely setting bits) → electronic signal stuff
2. · Data Link Layer – PDU = frame
   - reliable bitstream across one hop
3. · Network Layer – PDU = datagram
   - picks path across the network
4. · Transport Layer – PDU = segment
   error detection + correction, flow control
5. · Session Layer
   supports user connections
6. · Presentation Layer
   provides general solns to common requests
7. · Application Layer (HTTP: L7 header)
   provides access for user and application



L4 looks at network as a black box (end-to-end)

↳ its able to recover lost data if one node goes down

TCP/IP Protocol

Layers 1 and 2:
Network-to-host layer or
Network Interface layer

Ethernet owned by IEEE
· physical layer
· PDU = Frame

Layer 3: Internet Protocol (IP) → datagram
· routes
· picks the path

Layer 4: Transmission Control Protocol (TCP) segment
(END-to-END reliable bitstream)
· connection oriented
· reliable

or UDP (unreliable)

Layer 5: Application

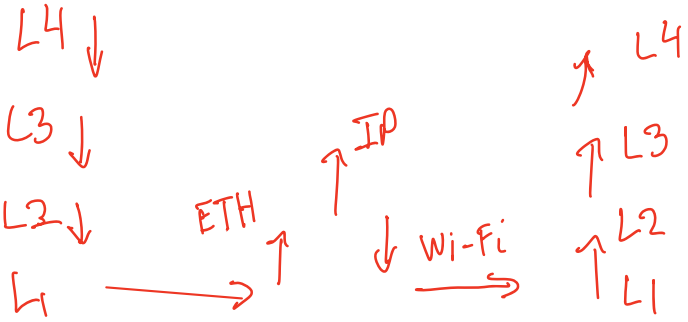
or

Layer 7 (7 layer)

HTTP
FTP
SMTP

IMAP
POP

802.11 = WiFi



Program 1

Files given : Contain packets w/ various headers

all start w/ ETH

ETH	IP	TCP
ETH	HTTP	
ETH		UDP

diff answers

14 bytes

6 bytes 6 bytes 2 bytes

dest MAC	source MAC	type	

learn how to process this header.